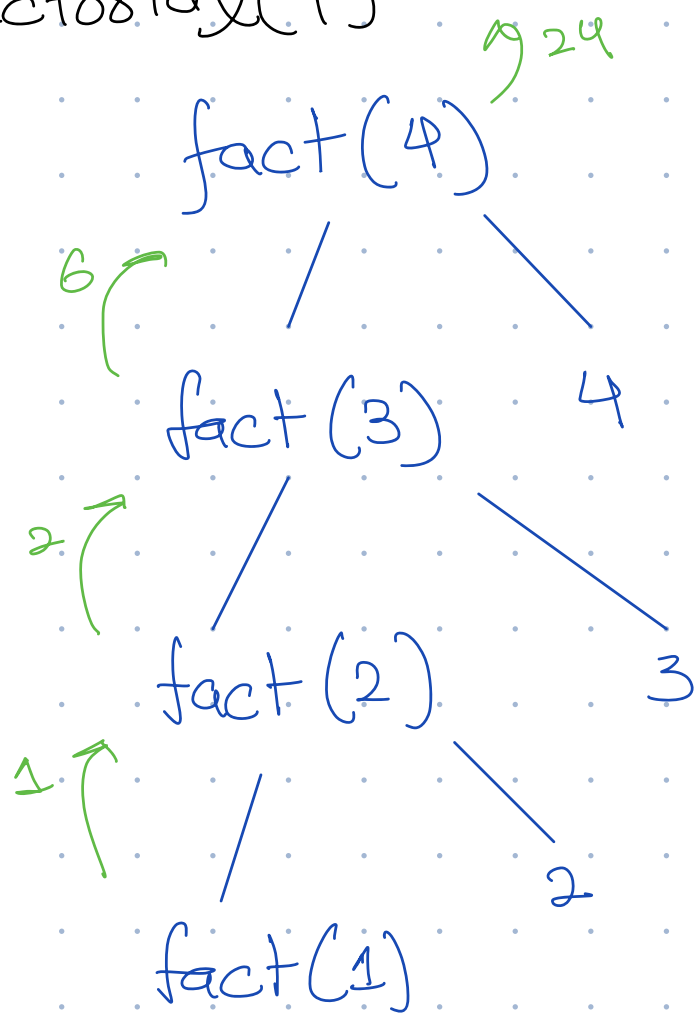
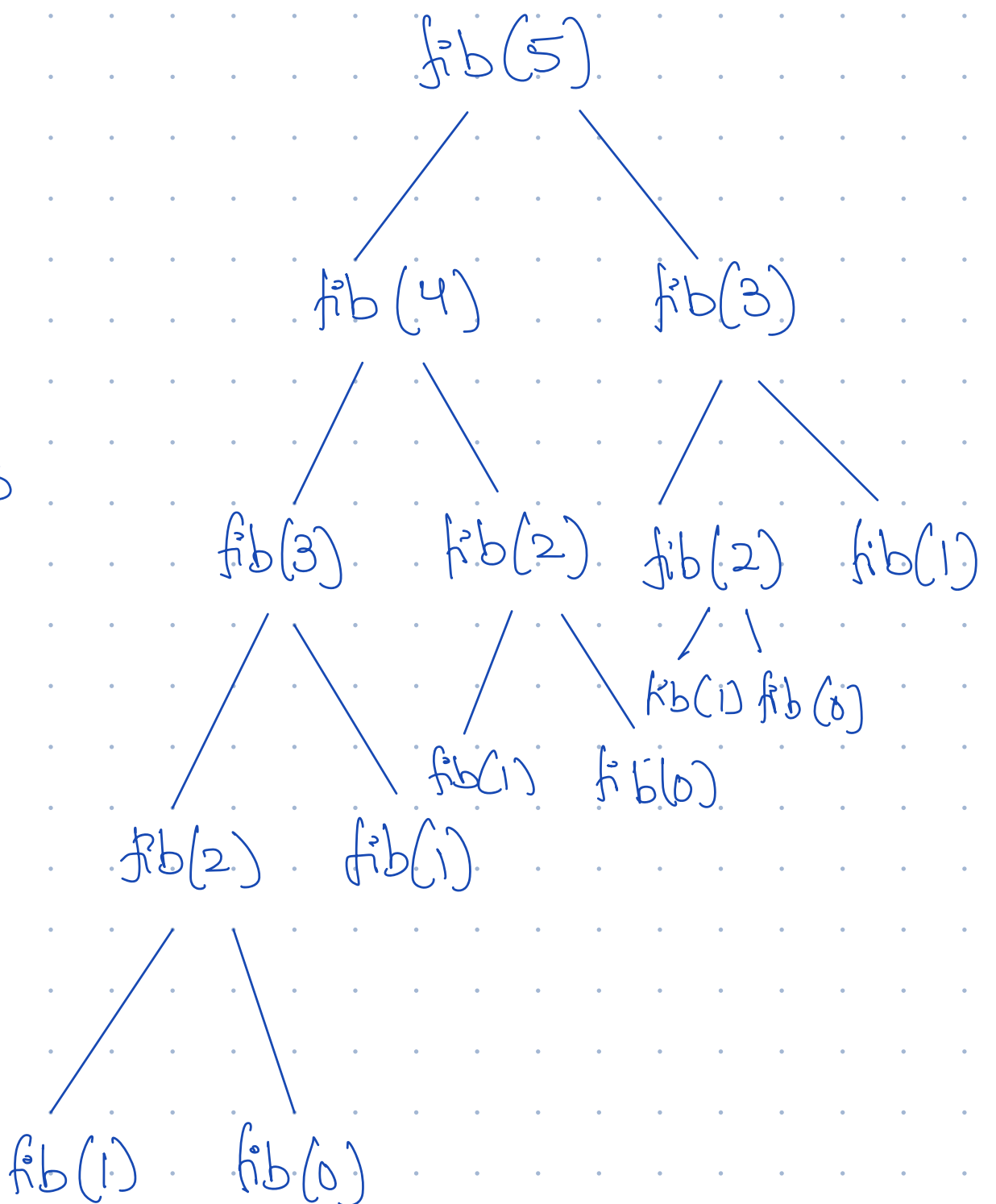


Factorial(4)



$O(n)$

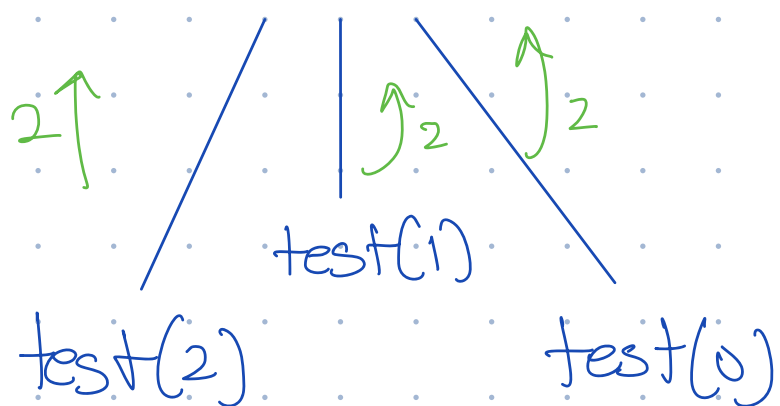
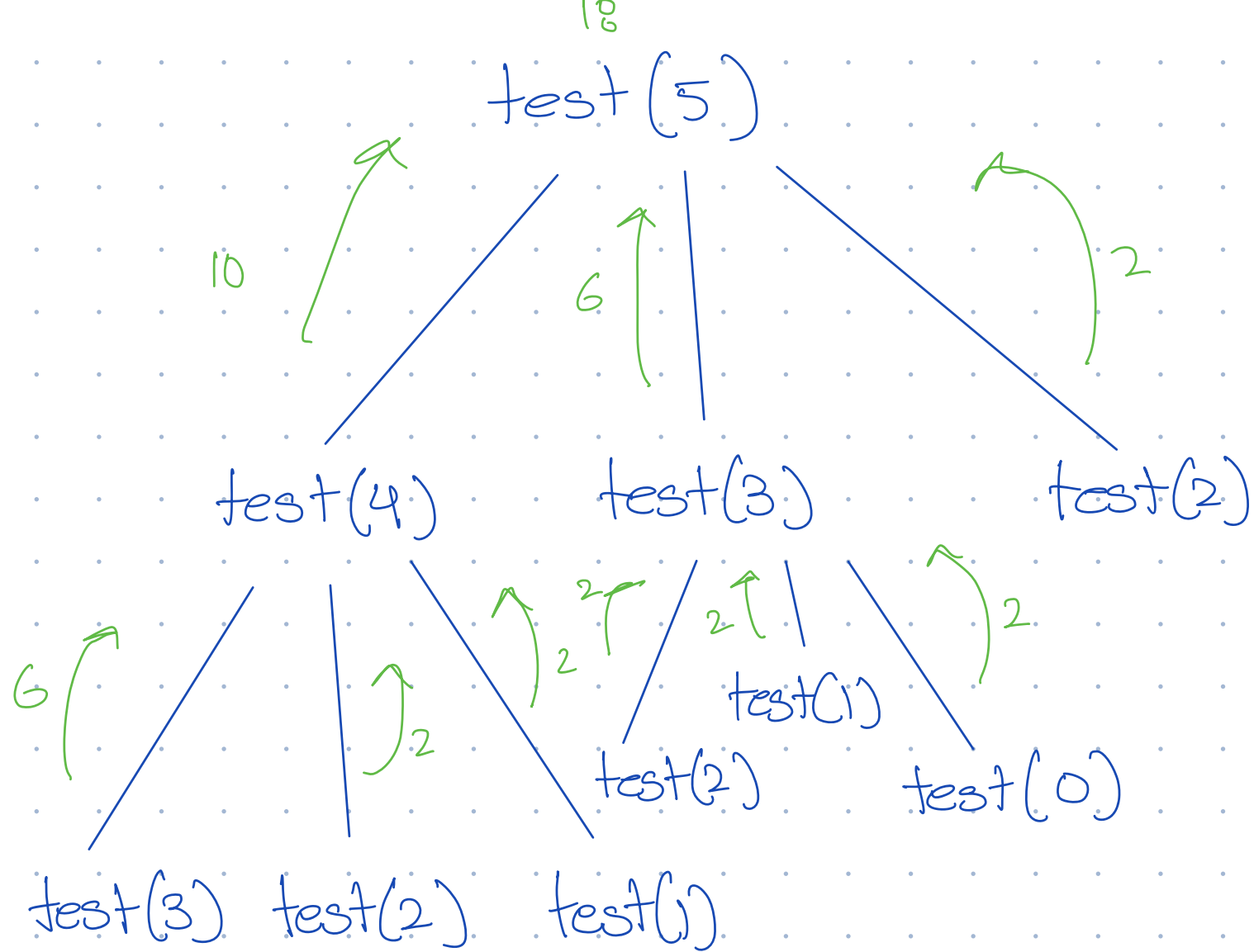
Fibonacci(5)



$O(2^n)$

1. Derive the complexity of following recursive code and what value will it return for test (5)?

```
int test (int n)
{
    if (n>2)
    {
        int c = 0;
        for (int i=1; i<=3; i++)
        {
            c+=test (n-i)
        }
        return c;
    }
    return 2;
}
```



Time complexity =  $3^n$

